

KEY U1L2 Mastery Check (Variables)

1. Choose the Best Variable Type

- a. A student's first name → **string**
- b. Number of pets you have → **integer**
- c. Price of a sandwich (\$5.75) → **float**
- d. Whether the class is in session → **boolean**

2. Variable Reassignment

```
points = 10
points = points + 5
print(points)
```

Answer: 15

3. Fix the Concatenation Error

Incorrect:

```
print("Age: " + age)
```

Correct version:

```
print("Age:", age)
```

OR

```
print("Age: " + str(age))
```

4. Predict and Explain

```
name = "Ava"
name = "Jordan"
print(name)
```

Answer: Jordan

Explanation: The variable `name` was reassigned to a new value. Python always prints the most recent assignment.

5. Converting an Integer to a Float

Convert 7 to float and print:

```
num = 7
num = float(num)
print(num)
```

Output: **7.0**

KEY U1L3 Mastery Check (Data Types)

1. Identify the Data Types

- a. "Chicago" → **string**
- b. 42 → **integer**
- c. 3.14 → **float**
- d. False → **boolean**
- e. "True" → **string**
- f. 18.0 → **float**
- g. -7 → **integer**
- h. "7" → **string**
- i. True → **boolean**
- j. 0 → **integer**

2. Fix the Error

Incorrect:

```
print("Height: " + 1.75)
```

Explanation: Can't concatenate string + float.

Corrected:

```
print("Height:", 1.75)
```

OR

```
print("Height: " + str(1.75))
```

3. Predict the Output

```
print(type("7"))
```

Output:

```
<class 'str'>
```

4. Valid Boolean Values

From the list: **True, False**

(Correct Python booleans must be capitalized.)

Answer: **True, False**

5. Type Conversion

Convert "29" into an integer and print type:

```
num = int("29")
```

```
print(type(num))
```

Output:

```
<class 'int'>
```